

ANDREA DE PISAPIA

Control & Automation Engineer

PROFILE

Control & Automation Engineer with strong expertise in real-time systems, advanced control strategies (MPC, nonlinear control), and embedded implementations.

Currently contributing to safety-critical control systems at Fusion for Energy, working on high-speed data acquisition and real-time communication architectures for ITER Gyrotron Plants.

Background in robotics (ROS, Gazebo), system modeling (MATLAB/Simulink), and real-time software, with a focus on intelligent and reliable control systems.

WORK EXPERIENCE

Embedded Systems Engineer (Trainee)

Fusion for Energy 

 Oct 2024 – Present  Barcelona, Spain

- Developed FPGA-based signal processing for high-speed ADC data acquisition in the **HotRIO control system** for ITER Gyrotron Plants.
- Integrated FPGA firmware with **CompactRIO**, optimizing real-time communication over fiber optics.
- Designed **diagnostic tools** and performed timing analysis, improving system reliability and performance.
- Simulated and tested VHDL designs using **ModelSim** for validation before deployment.
- Collaborated with **multidisciplinary teams**, participating in design reviews and technical discussions to enhance system capabilities.

PROJECTS

- **Advanced Control Systems (MPC, Robust & Nonlinear Control)** - Designed and validated Model Predictive Control (MPC, SMPC, RMPC) strategies under uncertainty (Monte Carlo, ARIMA), and applied nonlinear control techniques (Sliding Mode) to dynamic systems, achieving stable and robust performance.
- **Automation & Industrial Control Applications** - Developed control systems in Simulink for industrial processes, including asynchronous motor drives (scalar control) and thermo-hydraulic regulation with zero steady-state error.
- **Robotics & Autonomous Systems (ROS, Gazebo)** - Designed control algorithms and trajectory planning for SCARA robots, mobile platforms, and multi-DOF manipulators; implemented autonomous navigation using ArUco markers and ROS-based architectures.
- **Real-Time & Fault-Tolerant Systems** - Implemented real-time control loops and monitoring systems using FreeRTOS, including watchdog-based fault detection and encoder simulation for reliable system behavior.
- **Connected & Intelligent Systems** - Developed control strategies for autonomous vehicle platoons using V2X communication and designed multi-layer control architectures for energy-efficient systems.
- **System Modeling & Multidisciplinary Design** - Contributed to virtual prototyping and system-level optimization (e.g. ergonomic system design using RFLP methodology in automotive context).

CONTACT INFORMATION

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 Barcelona, ES

SKILLS

- **Control & Simulation:**
MATLAB, Simulink, Simscape
- **Embedded & Hardware:**
FPGA (VHDL, Lattice Diamond),
real-time communication systems
- **Real-Time Systems:**
FreeRTOS
- **Robotics:**
ROS, Gazebo
- **Programming:**
C, C++, Python
- **Other:**
CATIA, Xpress, Linux, Ubuntu

LANGUAGES

 Italian (Native Speaker)

 English (C1)

 Spanish (B1)

EDUCATION

Master of Automation Engineering and Robotics - 110/110

University of Napoli Federico II

 A.A: 2021/22 -> 2023/24

 Napoli, NA

 Thesis Title:
Development and Implementation of the
HotRIO Control System for ITER Gyrotron
Plants

Bachelor of Automation Engineering - 101/110

University of Napoli Federico II

 A.A: 2017/18 -> 2020/21 Napoli,

 NA

 Thesis Title:
Arduino Controlled Rover Project